SUPPORTING ADULT GUIDANCE

Changing the subject, harder cases

Algebraic Expressions and Identities · Lesson 6. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson extends changing the subject to formulas where the subject appears twice, as a square or under a root. In the main activity, learners write their own rearrangement one line at a time; each line is checked by substitution, so a step that changes the meaning is flagged at once (and a line where the subject vanishes, or is cancelled away, is caught). Progressive hints name the next move. They then spot faulty rearrangements and, in Creativity, hunt for self-inverse formulas of the form $y = (x + a)/(x + b)$.

Driving Question: *What if the subject appears twice, or as a square?*

The mathematics, in plain terms

Each step applies the same invertible operation to both sides, so the solution set is preserved — except that dividing by an expression requires it to be non-zero, and taking a square root requires choosing a sign. Hence conditions such as $y \neq 1$ or $r > 0$. For $y = (x + a)/(x - 1)$, the inverse is $x = (y + a)/(y - 1)$: the function is its own inverse.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The subject appears twice

Make x the subject of $y = (x + 2)/(x - 3)$.

1. Multiply both sides by $(x - 3)$.	$y(x - 3) = x + 2$
2. Expand the left side.	$xy - 3y = x + 2$
3. Collect the x terms on the left, the rest on the right.	$xy - x = 3y + 2$
4. Factorise: x appears once.	$x(y - 1) = 3y + 2$
5. Divide by $(y - 1)$, which needs $y \neq 1$.	$x = (3y + 2)/(y - 1)$

$x = (3y + 2)/(y - 1)$, for $y \neq 1$.

Example 2 — The subject is squared

Make r the subject of $A = \pi r^2$.

1. Divide both sides by π. $r^2 = A/\pi$
2. Undo the square with a square root. $r = \sqrt{(A/\pi)}$
3. r is a radius, a length, so take the positive root. $r > 0$
4. Check: if $r = 3$, $A = 9\pi$, and $\sqrt{(9\pi/\pi)} = 3$. ✓

$r = \sqrt{(A/\pi)}$.

Your own lines, checked

Learners are not given a fixed route: any valid sequence of lines is accepted, because each line is checked against the original formula by substitution. This keeps ownership with the learner while catching slips immediately. The key new move — collect and factorise — is named in the hints, and the stretch explains where conditions come from.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Once, then twice: three one-step rearrangements as a warm-up.	~5 min
Movement	The line-by-line balance: write each line; every line is checked. Stretch: where conditions come from.	~18 min
Reflection	Spot the faulty rearrangement.	~8 min
Creativity	Turn it inside out: find the self-inverse formulas.	~9 min
Rest	A tangled letter, gathered and set free.	~2 min
Nutrition	Intellectual: persistence with structure.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Trying to divide by x when x appears on both sides.
Ask: "Can you get every x term on one side first?"

Sign slip when moving a term: $xy - 3y = x + 2 \rightarrow xy - x = 2 - 3y$.
Ask: "What do you add to both sides to move $-3y$?"

Dividing only part of a side: $x = 7 + y/3$.
Ask: "Did you divide every term on that side?"

Halving instead of square rooting.
Ask: "What undoes squaring?"

Forgetting the condition after dividing by an expression.
Ask: "Could that expression ever be zero?"

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line
Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson's Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: changing the subject when it appears once, and taking out a common factor. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner, rearranging $y = (x + 1)/(x - 4)$, writes $xy - x = 1 - 4y$. The aim is a question that lets them find the slip.

- "What was the line before? What did you do to both sides?"

- “Put x = 5 into the original: what is y? Does your line hold?”
- “When $-4y$ moves across, what happens to its sign?”
- “So what should the right side be?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Must learners follow the same steps as the hints?
No. Any valid line is accepted; the hints are one route among several.

Is $x = (-3y - 2)/(1 - y)$ accepted?
Yes: it is the same formula, checked by value.

Is the proof required?
It is a stretch. The secure core is collect, factorise, divide, and check.

What comes next?
Lesson 7: algebraic proof.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.
Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.

Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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